Assignment 1 MAST90125: Bayesian Statistical Learning

Due: 11:59pm Sunday 23 August 2026

There are places in this assignment where R code will be required. Therefore set the random seed so your assignment result is reproducible.

```
set.seed(1628985)  #Please change random seed to your student id number.
```

Please save this R markdown document and write your answers in it. Between your answer to each question, ensure there is sufficient space for marker comments by using the command

```
\newpage
```

Question One (5 marks)

Suppose two data observations y_1 and y_2 are independently generated from a data generative model with pdf $p(y|\theta)$. Let $p(\theta)$ be a prior pdf of θ, and $p(\theta|y_1, y_2)$, $p(\theta|y_1)$ and $p(\theta|y_2)$ be the posterior pdf's conditional on (y_1, y_2), y_1 and y_2, respectively.

a) Is it valid to estimate $p(\theta|y_1)$ and $p(\theta|y_2)$ seperately, and then obtain the posterior $p(\theta|y_1, y_2)$ as $p(\theta|y_1, y_2) = p(\theta|y_1)p(\theta|y_2)$? Justify your answer using Bayes Theorem.

No, this is not valid.

By Bayes Theorem:

$$p(\theta|y_1, y_2) = \frac{p(y_1, y_2|\theta)p(\theta)}{p(y_1, y_2)}$$

Since y_1 and y_2 are independent:

$$p(\theta|y_1, y_2) = \frac{p(y_1|\theta)p(y_2|\theta)p(\theta)}{p(y_1)p(y_2)}$$

Also:

$$p(\theta|y_1) = \frac{p(y_1|\theta)p(\theta)}{p(y_1)}$$

$$p(\theta|y_2) = \frac{p(y_2|\theta)p(\theta)}{p(y_2)}$$

Therefore:

$$p(\theta|y_1)p(\theta|y_2) = \frac{p(y_1|\theta)p(y_2|\theta)p(\theta)^2}{p(y_1)p(y_2)}$$

Comparing this to $p(\theta|y_1, y_2)$:

$$p(\theta|y_1, y_2) = \frac{p(\theta|y_1)p(\theta|y_2)}{p(\theta)}$$

So $p(\theta|y_1, y_2) \neq p(\theta|y_1)p(\theta|y_2)$ in general.

b) If you really want to use $p(\theta|y_1)$ and $p(\theta|y_2)$ to obtain $p(\theta|y_1, y_2)$, what else is needed in order to do so? Provide the answer and justification.

We need the prior $p(\theta)$.

From the derivation above:

$$p(\theta|y_1, y_2) = \frac{p(\theta|y_1)p(\theta|y_2)}{p(\theta)}$$

Therefore, what we need is $p(\theta)$, since $p(\theta|y_1)$ and $p(\theta|y_2)$ is known.

Question Two (5 marks)

Medical researchers wish to study the performance of a diagnostic test. Prior studies suggest the underlying probability of disease (event A) is p_A. To determine the effectiveness of the diagnostic test (event B = testing positive), a case-control study was undertaken. Both cases and controls were continuously added to the study to be tested until d_1 cases tested positive, and d_2 controls tested negative, respectively. Here d_1 and d_2 are pre-specified.

a) Let observation $n_{\bar{B}|A}$ be the number of cases testing negative, and observation $n_{B|\bar{A}}$ be the number of controls testing positive. Identify the likelihood of $n_{\bar{B}|A}$ and $n_{B|\bar{A}}$, respectively, using the corresponding probability mass functions (pmf's) conditional on appropriate parameter(s). Let $p = P(B|A)$ be the sensitivity (probability of testing positive given disease) and $q = P(B|\bar{A})$ be the false positive rate (probability of testing positive given no disease).

Sensitivity

- d_1 means testing positive
- The number of cases testing negative $n_{\bar{B}|A}$ follows a Negative Binomial distribution with parameters d_1 and p_1 where $p_1 = P(\text{testing negative}|\text{disease}) = 1 - p$

So:

$$P(n_{\bar{B}|A}|d_1, p) = \binom{n_{\bar{B}|A} + d_1 - 1}{d_1 - 1}(1-p)^{n_{\bar{B}|A}}p^{d_1}$$

False positive

- d_2 means testing negative (successes)
- The number of controls testing positive $n_{B|\bar{A}}$ also follows a Negative Binomial distribution with parameters d_2 and q (where $q = P(\text{testing positive}|\text{no disease})$)

So:

$$P(n_{B|\bar{A}}|d_2, q) = \binom{n_{B|\bar{A}} + d_2 - 1}{d_2 - 1}q^{n_{B|\bar{A}}}(1-q)^{d_2}$$

b) Identify a suitable conjugate prior for each parameter used in a).
For the Negative Binomial parameter $1 - p$ (or q), the conjugate prior is the **Beta distribution**.

For $1 - p$:

$$p \sim \text{Beta}(\alpha_1, \beta_1)$$

where α_1 and β_1 are hyperparameters.

For q:

$$q \sim \text{Beta}(\alpha_2, \beta_2)$$

where α_2 and β_2 are hyperparameters.

c) Determine the posterior distribution for each parameter used in a). For the parameter $(1-p)$:

- Likelihood: $P(n_{\bar{B}|A}|d_1, 1-p) \propto (1-p)^{n_{\bar{B}|A}} p^{d_1} = (1-p)^{n_{\bar{B}|A}} (1-(1-p))^{d_1}$
- Prior: $(1-p) \sim \text{Beta}(\alpha_1, \beta_1)$
- Posterior: $(1-p)|n_{\bar{B}|A}, d_1 \sim \text{Beta}(\alpha_1 + n_{\bar{B}|A}, \beta_1 + d_1)$

Therefore:

$$p|n_{\bar{B}|A}, d_1 \sim \text{Beta}(\beta_1 + d_1, \alpha_1 + n_{\bar{B}|A})$$

For the parameter q:

- Likelihood: $P(n_{B|\bar{A}}|d_2, q) \propto q^{n_{B|\bar{A}}} (1-q)^{d_2}$
- Prior: $q \sim \text{Beta}(\alpha_2, \beta_2)$
- Posterior: $q|n_{B|\bar{A}}, d_2 \sim \text{Beta}(\alpha_2 + n_{B|\bar{A}}, \beta_2 + d_2)$

Question Three (5 marks)

As part of an investigation into traffic flows, a study was proposed to count the number of vehicles passing through an intersection each minute between 5 pm and 6 pm for one week. The researchers have decided to assume the resulting counts are *i.i.d.* both within and between days.

a) Specify an appropriate likelihood function for the data in the above situation. Also derive the Jeffreys' prior for the involved parameter.

The appropriate likelihood is the **Poisson distribution**:

$$p(y_1, \ldots, y_n|\lambda) = \prod_{i=1}^{n} \frac{\lambda^{y_i} e^{-\lambda}}{y_i!} = \frac{\lambda^{\sum y_i} e^{-n\lambda}}{\prod y_i!}$$

The Fisher information for λ:

$$I(\lambda) = -E\left[\frac{\partial^2}{\partial \lambda^2} \log p(y|\lambda)\right]$$

For Poisson:

$$\log p(y|\lambda) = y \log \lambda - \lambda - \log y!$$

$$\frac{\partial}{\partial \lambda} \log p(y|\lambda) = \frac{y}{\lambda} - 1$$

$$\frac{\partial^2}{\partial \lambda^2} \log p(y|\lambda) = -\frac{y}{\lambda^2}$$

$$I(\lambda) = -E\left[-\frac{y}{\lambda^2}\right] = \frac{E[y]}{\lambda^2} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}$$

Jeffreys' prior is:

$$p_J(\lambda) \propto \sqrt{I(\lambda)} = \frac{1}{\sqrt{\lambda}}$$

b) In this case, is Jeffreys' prior improper? Justify your answer.

Yes, Jeffreys' prior is improper.

The prior $p_J(\lambda) \propto \lambda^{-1/2}$ for $\lambda > 0$ is not a proper probability density because:

$$\int_0^\infty \lambda^{-1/2} d\lambda = \infty$$

This means the prior cannot be normalized to integrate to 1.

c) In this case does Jeffreys' prior satisfy the criterion:

$$\text{Posterior} \propto \text{Likelihood?}$$

Justify your answer.

No

With Jeffreys' prior $p_J(\lambda) \propto \lambda^{-1/2}$:

$$p(\lambda|y_1, \ldots, y_n) \propto \lambda^{\sum y_i} e^{-n\lambda} \times \lambda^{-1/2} = \lambda^{\sum y_i - 1/2} e^{-n\lambda}$$

For Posterior $\propto$ Likelihood, we need $p(\lambda|y_1, \ldots, y_n) \propto \lambda^{\sum y_i} e^{-n\lambda}$, which would require a flat prior $p(\lambda) \propto 1$. The Jeffreys' prior is not flat, so the posterior differs from the likelihood.

Question Four (11 marks)

To comply with food labelling regulations, a manufacturer of certain product (e.g. chocolate in packs) must prove that 99% of its produced packs are each within 7g of the stated weight $\mu = 250$g.

The manufacturer knows that the average weight of its produced packs equals the stated weight. It also knows that the machinery has been designed such that the respective weights of the products are normally distributed with mean equal to the stated weight and an unknown variance σ^2. To test the machinery is working to the specification, the company randomly selected $n = 100$ packs from the production lines and calculated the residual $(y_i - \mu)$ weight. They reported the sum of squared residuals SSR $= \sum_{i=1}^{n}(y_i - \mu)^2$ was 572.78.

a) Let $\theta = \sigma^2$ be the parameter on which you wish to make inference. By appropriate manipulation of the likelihood of $(y_1, \ldots, y_n)$, demonstrate that SSR is the sufficient statistic for θ.

Given $y_i \sim N(\mu, \sigma^2)$ with known $\mu = 250$:

$$
\begin{aligned}
L(\theta|y_1, \ldots, y_n) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\theta}} \exp\left(-\frac{(y_i - \mu)^2}{2\theta}\right) \\
&= (2\pi\theta)^{-n/2} \exp\left(-\frac{1}{2\theta}\sum_{i=1}^{n}(y_i - \mu)^2\right) \\
&= (2\pi\theta)^{-n/2} \exp\left(-\frac{SSR}{2\theta}\right)
\end{aligned}
$$

By the Factorization Theorem, since the likelihood depends on the data only through $SSR = \sum(y_i - \mu)^2$, SSR is a sufficient statistic for θ.

b) Choose an appropriate one-to-one transformation $f(\theta)$ so that $f(\theta)$ represents the precision parameter in the likelihood of $(y_1, \ldots, y_n)$. Write down a conjugate prior for the new parameter $\tau = f(\theta)$.

Let $\tau = f(\theta) = 1/\theta = 1/\sigma^2$.

The likelihood in terms of τ:

$$L(\tau|y_1, \ldots, y_n) \propto \tau^{n/2} \exp\left(-\frac{\tau}{2}\sum(y_i - \mu)^2\right)$$

This is proportional to a Gamma distribution in τ:

$$\tau \sim \text{Ga}\left(\frac{n}{2}, \frac{SSR}{2}\right)$$

The conjugate prior for τ is:

$$\tau \sim \text{Ga}(\alpha_0, \beta_0)$$

with density:

$$p(\tau) = \frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)}\tau^{\alpha_0 - 1}e^{-\beta_0\tau}$$

c) Determine the posterior distribution $p(f(\theta)|y_1, \ldots, y_n)$. Substituting 1 for the hyper-parameters of the prior in b), determine the 95% central credible interval for θ.

With prior $\tau \sim \text{Ga}(\alpha_0, \beta_0)$:

$$p(\tau|y_1, \ldots, y_n) \propto \tau^{n/2} \exp\left(-\frac{\tau}{2} SSR\right) \times \tau^{\alpha_0 - 1} e^{-\beta_0 \tau}$$
$$= \tau^{\alpha_0 + n/2 - 1} \exp\left(-\left(\beta_0 + \frac{SSR}{2}\right)\tau\right)$$

So:

$$\tau|y_1, \ldots, y_n \sim \text{Ga}\left(\alpha_0 + \frac{n}{2}, \beta_0 + \frac{SSR}{2}\right)$$

With $\alpha_0 = 1, \beta_0 = 1$ and $n = 100, SSR = 572.78$:

$$\tau|data \sim \text{Ga}\left(1 + 50, 1 + \frac{572.78}{2}\right) = \text{Ga}(51, 287.39)$$

For $\theta = 1/\tau$: The posterior distribution of θ follows an **Inverse-Gamma** distribution:

$$\theta|data \sim \text{Inv-Ga}(51, 287.39)$$

Compute the interval:

```
# 95% central credible interval for theta
alpha_post <- 51
beta_post <- 287.39

# For tau ~ Gamma(alpha, beta)
tau_lower <- qgamma(0.025, shape = alpha_post, rate = beta_post)
tau_upper <- qgamma(0.975, shape = alpha_post, rate = beta_post)

# For theta = 1/tau
theta_CI_lower <- 1/tau_upper
theta_CI_upper <- 1/tau_lower

theta_CI_lower
```

```
## [1] 4.35976
```

```
theta_CI_upper
```

```
## [1] 7.568302
```

The HPD interval is approximately [4.35976, 7.568302]

d) Using the posterior distribution in c), determine the 95% highest posterior density interval for θ, and compare it to the 95% central credible interval.

```r
# Function to find HPD interval
find_hpd <- function(alpha, beta, conf = 0.95) {

  # Function to minimize: length of interval
  objective <- function(lower_prob) {
    upper_prob <- lower_prob + conf
    lower_tau <- qgamma(lower_prob, shape = alpha, rate = beta)
    upper_tau <- qgamma(upper_prob, shape = alpha, rate = beta)

    # Convert to theta scale
    lower_theta <- 1/upper_tau
    upper_theta <- 1/lower_tau

    # Check equal density condition
    density_lower <- dgamma(lower_tau, shape = alpha, rate = beta) / (lower_tau^2)
    density_upper <- dgamma(upper_tau, shape = alpha, rate = beta) / (upper_tau^2)

    return((upper_theta - lower_theta) + 1000 * abs(density_lower - density_upper))
  }

  # Optimize
  result <- optimize(objective, interval = c(0, 1 - conf))
  lower_prob <- result$minimum
  upper_prob <- lower_prob + conf

  lower_tau <- qgamma(lower_prob, shape = alpha, rate = beta)
  upper_tau <- qgamma(upper_prob, shape = alpha, rate = beta)

  return(c(1/upper_tau, 1/lower_tau))
}

hpd_interval <- find_hpd(alpha_post, beta_post)
hpd_interval
```

```
## [1] 4.501352 8.207328
```

e) Do you believe, based on the above posterior inference, that the machinery used by this manufacturer satisfies the requirement that 99% of its products with the stated weights 250g per pack are within 7g of the stated weight?

Given $y_i \sim N(\mu, \sigma^2)$:

$$P(|y_i - \mu| \leq 7) = P(-7/\sigma \leq Z \leq 7/\sigma) = 2\Phi(7/\sigma) - 1$$

We require $2\Phi(7/\sigma) - 1 \geq 0.99$, which means $\Phi(7/\sigma) \geq 0.995$, so $7/\sigma \geq 2.576$, or $\sigma \leq 7/2.576 \approx 2.718$.

Therefore we need $\theta = \sigma^2 \leq 7.39$.

The 95% credible interval for θ is approximately $[4.35976, 7.568302]$. ince the **lower bound of the credible interval (4.36) is greater than 2.718**, the data provide strong evidence that σ is actually larger than the required threshold.

Based on the posterior inference, we cannot confidently believe that the machinery satisfies the requirement that 99% of products are within 7g of the stated weight.

Question Five (9 marks)

There are $N = 112$ students enrolled in the Master of Science in the School of Mathematics and Statistics. At the end of the semester, $n = 35$ responses were received to an experience survey sent to Master's students. Among the questions asked was whether they felt adequate support was provided by the School. In $y = 17$ of the responses, the answer was yes.

a) Regarding the $n = 35$ students responding as the sample obtained from sampling with replacement (approximately), write down an appropriate single parameter distribution $p(y|\theta)$ corresponding to the likelihood of y.

For $n = 35$ students responding, with $y = 17$ answering "yes", we use the Binomial distribution:

$$p(y|\theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}$$

where θ is the true proportion of all 112 Master's students who feel adequate support is provided.

b) Identify a prior distribution, $p(\theta)$ that is conjugate with the likelihood chosen in a).

The conjugate prior for the Binomial parameter θ is the **Beta distribution**:

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$p(\theta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

c) Determine the posterior distribution $p(\theta|y)$ based on your choices of likelihood and prior in a) and b). Using the Beta prior $\theta \sim \text{Beta}(\alpha, \beta)$ and Binomial likelihood:

$$\begin{aligned} p(\theta|y) &\propto \theta^y(1-\theta)^{n-y} \times \theta^{\alpha-1}(1-\theta)^{\beta-1} \\ &= \theta^{\alpha+y-1}(1-\theta)^{\beta+n-y-1} \end{aligned}$$

So:

$$\theta|y \sim \text{Beta}(\alpha + y, \beta + n - y)$$

With the default choice of $\alpha = 1, \beta = 1$:

$$\theta|y = 17, n = 35 \sim \text{Beta}(18, 19)$$

d) Determine the posterior predictive distribution $p(\tilde{y}|y)$.

The posterior predictive distribution for a new observation $\tilde{y}$:

$$p(\tilde{y}|y) = \int p(\tilde{y}|\theta)p(\theta|y)d\theta$$

For Binomial-Beta:

$$p(\tilde{y}|y) = \int \binom{m}{\tilde{y}}\theta^{\tilde{y}}(1-\theta)^{m-\tilde{y}} \times \frac{\Gamma(\alpha+\beta+n)}{\Gamma(\alpha+y)\Gamma(\beta+n-y)}\theta^{\alpha+y-1}(1-\theta)^{\beta+n-y-1}d\theta$$

$$= \binom{m}{\tilde{y}}\frac{\Gamma(\alpha+\beta+n)}{\Gamma(\alpha+y)\Gamma(\beta+n-y)}\int \theta^{\alpha+y+\tilde{y}-1}(1-\theta)^{\beta+n-y+m-\tilde{y}-1}d\theta$$

$$= \binom{m}{\tilde{y}}\frac{\Gamma(\alpha+\beta+n)}{\Gamma(\alpha+y)\Gamma(\beta+n-y)}\frac{\Gamma(\alpha+y+\tilde{y})\Gamma(\beta+n-y+m-\tilde{y})}{\Gamma(\alpha+\beta+n+m)}$$

This is the Beta-Binomial distribution.

With $\alpha = 1, \beta = 1, y = 17, n = 35$, for a future survey of m students:

$$p(\tilde{y}|y) = \binom{m}{\tilde{y}}\frac{\Gamma(37)}{\Gamma(18)\Gamma(19)}\frac{\Gamma(18+\tilde{y})\Gamma(19+m-\tilde{y})}{\Gamma(37+m)}$$

Some useful density functions

- Normal distribution: $y \sim N(\mu, \sigma^2)$

$$p(y|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$$

- Binomial distribution: $y \sim \text{Bin}(n, p)$

$$\Pr(y|n, p) = \binom{n}{y} p^y (1-p)^{n-y}$$

- Poisson distribution: $y \sim \text{Poi}(\lambda)$

$$\Pr(y|\lambda) = \frac{\lambda^y}{y!} e^{-\lambda}$$

- Negative binomial distribution: $y \sim \text{NBin}(d, p)$ where y is the number of Bernoulli failures before the d-th success and p is the probability of a Bernoulli failure

$$\Pr(y|d, p) = \binom{y+d-1}{d-1} p^y (1-p)^d$$

- Gamma distribution: $y \sim \text{Ga}(\alpha, \beta)$

$$p(y|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}$$

- Beta distribution: $y \sim \text{Beta}(\alpha, \beta)$

$$p(y|\alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} y^{\alpha-1} (1-y)^{\beta-1}$$